

Refutation of Conjecture 00000001065: the Nikodym–Kakeya difference in $\mathbb{F}_q^2$

Abstract

Conjecture 00000001065 asserts that, in the plane over the finite field $\mathbb{F}_q$, the minimal size of a Nikodym set exceeds the minimal size of a Kakeya set by exactly $q - 1$. Under the definitions recorded in the conjecture itself, this is false already for $q = 2$, and in fact the difference is not $q - 1$ for $q = 2, 3, 4$. We give an exhaustive verification for $q = 2, 3, 4$ together with a complete machine-checked proof of the decisive $q = 2$ case in core Lean 4.

1 Definitions

Let q be a prime power and let $\mathbb{F}_q$ be the field with q elements. A *line* in the affine plane $\mathbb{F}_q^2$ has exactly q points. The plane has $q(q + 1)$ lines, partitioned into $q + 1$ *directions* (the points of the projective line $\mathbb{P}^1(\mathbb{F}_q)$), each direction containing q parallel lines.

Definition 1 (Kakeya set). A subset $K \subseteq \mathbb{F}_q^2$ is a *Kakeya set* if for every one of the $q + 1$ directions there is a line of that direction that is fully contained in K .

Definition 2 (Nikodym set). A subset $N \subseteq \mathbb{F}_q^2$ is a *Nikodym set* if for every point $p \in \mathbb{F}_q^2$ there is a line L through p such that

$$L \setminus \{p\} \subseteq N.$$

In words: through every point there is a line whose *other* points all lie in N ; the point p itself is allowed to be exceptional.

These are the definitions stated in Conjecture 00000001065 (“a Nikodym set: through each point, a line almost all of whose points lie in the set”). Note that in Definition 2 the quantifier runs over *all* points of the plane, including those outside N .

Definition 3 (Nikodym–Kakeya difference). Write $\kappa(q)$ for the minimal cardinality of a Kakeya set in $\mathbb{F}_q^2$ and $\nu(q)$ for the minimal cardinality of a Nikodym set. The *Nikodym–Kakeya difference* is $\nu(q) - \kappa(q)$.

Conjecture 00000001065. *The minimal size of a Nikodym set exceeds that of a Kakeya set by exactly $q - 1$; that is, $\nu(q) - \kappa(q) = q - 1$.*

2 The conjecture is false

Theorem 1. *Under Definitions 1 and 2,*

$$\nu(2) = 2, \quad \kappa(2) = 3, \quad \nu(2) - \kappa(2) = -1 \neq 1 = q - 1.$$

Consequently Conjecture 00000001065 is false.

The rest of the paper records the exhaustive computations that establish this and the analogous failures for $q = 3, 4$, and discusses the strongest possible objection.

3 Exhaustive computation

For each $q \in \{2, 3, 4\}$ we enumerated *all* 2^{q^2} subsets of $\mathbb{F}_q^2$ (16, 512, and 65 536 subsets respectively), tested Definition 1 and Definition 2 directly, and recorded the least cardinality at which each property occurs. Arithmetic in $\mathbb{F}_4$ was carried out with the irreducible polynomial $x^2 + x + 1$, so that the four elements are $0, 1, x, x + 1$ with $x^2 = x + 1$.

3.1 The case $q = 2$

Index the four points of $\mathbb{F}_2^2$ row-major as

$$(0, 0) \rightarrow 0, \quad (0, 1) \rightarrow 1, \quad (1, 0) \rightarrow 2, \quad (1, 1) \rightarrow 3.$$

The six lines are

$$\{0, 2\}, \{1, 3\} \quad (\text{horizontal}), \quad \{0, 1\}, \{2, 3\} \quad (\text{vertical}), \quad \{0, 3\}, \{1, 2\} \quad (\text{diagonal}),$$

one pair for each of the $3 = q + 1$ directions.

- **Takeya minimum is 3.** No set with fewer than three points can meet all three directions: a set of at most two points lies on at most one line, hence realises at most one direction. A triangle such as $\{(0, 0), (0, 1), (1, 0)\}$ contains the horizontal line $\{0, 2\}$, the vertical line $\{0, 1\}$ and the diagonal line $\{1, 2\}$, so it is Takeya. Hence $\kappa(2) = 3$.
- **Nikodym minimum is 2.** The two-point set $N = \{(0, 0), (0, 1)\}$ is Nikodym. Indeed, for the point $(0, 0)$ the vertical line $\{(0, 0), (0, 1)\}$ has its other point in N ; for $(0, 1)$ the same line works; for $(1, 0)$ the diagonal line $\{(1, 0), (0, 1)\}$ has its other point in N ; and for $(1, 1)$ the horizontal line $\{(1, 1), (0, 1)\}$ has its other point in N . A set of size 0 or 1 is never Nikodym, since for a point p outside N (or for p the unique point of N) every line through p has an other point outside N . Hence $\nu(2) = 2$.

Therefore $\nu(2) - \kappa(2) = 2 - 3 = -1$, whereas $q - 1 = 1$.

3.2 The case $q = 3$

There are $4 = q + 1$ directions and 9 points, hence 12 lines. Exhaustive enumeration of the 512 subsets gives

$$\kappa(3) = 7, \quad \nu(3) = 5, \quad \nu(3) - \kappa(3) = -2 \neq 2 = q - 1.$$

3.3 The case $q = 4$

Here $q = 4$ is a genuine prime power, not a prime; we used the field $\mathbb{F}_4 = \{0, 1, x, x + 1\}$ with $x^2 = x + 1$. There are 5 directions and 20 lines. Exhaustive enumeration of all 65 536 subsets gives

$$\kappa(4) = 10, \quad \nu(4) = 10, \quad \nu(4) - \kappa(4) = 0 \neq 3 = q - 1.$$

4 Results

The differences are negative or zero, not $q - 1$: the Nikodym minimum is *below* the Takeya minimum for $q = 2, 3$ and equal to it for $q = 4$.

q	min Keakeya $\kappa(q)$	min Nikodym $\nu(q)$	$\nu(q) - \kappa(q)$	claimed $q - 1$	verdict
2	3	2	-1	1	differs
3	7	5	-2	2	differs
4	10	10	0	3	differs

Table 1: Exhaustive minima in $\mathbb{F}_q^2$. The difference is negative (and for $q = 4$ zero), never $q - 1$; the conjecture fails in every computed case.

5 Sanity checks

The computed Keakeya minima agree with the known formula

$$\kappa(q) = \begin{cases} \frac{q(q+1)}{2}, & q \text{ even,} \\ \frac{q(q+1)}{2} + \frac{q-1}{2}, & q \text{ odd,} \end{cases}$$

giving $\kappa(2) = 3$, $\kappa(3) = 7$, $\kappa(4) = 10$, which match the enumeration. This validates the search independently of the Nikodym computation.

6 The strongest objection: the “whole line” reading

The definition of a Nikodym set is often strengthened by replacing “almost all” by “all”: one demands a *whole* line through every point contained in N , i.e. for every p there is a line L through p with $L \subseteq N$. Under this reading the minimum jumps to q^2 : if $p \notin N$, then the line L through p guaranteed by the definition satisfies $p \in L \subseteq N$, a contradiction; hence no point can lie outside N , so $N = \mathbb{F}_q^2$ and $\min = q^2$.

It is worth noting where the claimed formula $q - 1$ may have come from. Under the strengthened reading one has

$$q^2 - \kappa(q) = \frac{q(q+1)}{2} - \frac{q-1}{2} \text{ or } \frac{q(q+1)}{2}$$

and this coincidentally equals $q - 1$ for the two smallest cases:

$$q = 2 : \quad 4 - 3 = 1 = q - 1, \quad q = 3 : \quad 9 - 7 = 2 = q - 1.$$

But the coincidence already fails at the very next case:

$$q = 4 : \quad 16 - 10 = 6 \neq 3 = q - 1.$$

So even under the strengthened “whole line” reading the formula $q - 1$ survives only for $q = 2, 3$ and fails from $q = 4$ onward. Under the wording actually stated in Conjecture 00000001065, where the distinguished point is allowed to be exceptional, the refutation is immediate and holds already at $q = 2$.

Remark 1. The value q^2 is the trivial upper bound for a Nikodym set under the strengthened reading; the point of the computation above is only that the numerical coincidence $q^2 - \kappa(q) = q - 1$ at $q = 2, 3$ does not persist.

7 Conclusion

Under the definitions stated in Conjecture 00000001065, the Nikodym–Kakeya difference is *not* $q - 1$: for $q = 2$ it is -1 , for $q = 3$ it is -2 , and for $q = 4$ it is 0 . The smallest witness is the two-point set $\{(0, 0), (0, 1)\} \subseteq \mathbb{F}_2^2$, which is Nikodym, while every Kakeya set in $\mathbb{F}_2^2$ needs at least three points. The conjecture is therefore **false**. The machine-checked Lean 4 development accompanying this note formalises the $q = 2$ disproof completely.